

OLYMPIA, WA, USA

WMO: 727920

Lat: 46.973N

Lon: 122.903W

Elev: 57

StdP: 100.64

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24227

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.1	-4.1	-9.9	1.6	-2.9	-7.1	2.1	-2.6	10.9	8.9	9.3	7.6	0.5	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	31.0	18.7	28.8	18.1	26.7	17.4	19.8	29.1	18.8	27.3	17.9	25.4	3.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.8	21.6	15.6	11.1	20.8	14.8	10.6	20.1	56.4	29.1	53.1	27.1	50.3	25.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.1	-10.3	35.0	2.9	2.2	-12.3	36.6	-14.0	37.9	-15.6	39.1	-17.7	40.7		
(5)			-10.7	21.5	2.9	1.1	-12.8	22.2	-14.5	22.9	-16.1	23.5	-18.2	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.4	4.8	6.9	9.1	12.5	15.1	18.1	18.0	15.2	10.3	6.4	3.9
	DBStd	5.83	3.39	3.01	2.70	2.80	3.01	2.85	2.71	2.36	2.55	2.69	3.36	3.37
	HDD10.0	816	174	146	103	49	10	0	0	0	1	29	115	190
	HDD18.3	2955	431	378	355	278	182	104	38	36	99	248	358	447
	CDD10.0	971	1	1	6	22	87	154	249	248	155	38	7	1
	CDD18.3	68	0	0	0	0	1	8	29	26	4	0	0	0
	CDH23.3	1416	0	0	1	13	77	191	528	464	142	1	0	0
	CDH26.7	463	0	0	0	2	16	63	197	152	33	0	0	0
(14) Wind	WSAvg	2.4	2.5	2.5	2.8	2.7	2.5	2.5	2.3	2.1	2.0	2.3	2.6	2.5
(15) Precipitation	PrecAvg	1255	205	139	121	88	52	41	18	29	51	106	199	205
	PrecMax	1692	384	394	254	198	139	95	76	138	193	272	383	364
	PrecMin	890	7	6	12	32	5	4	0	0	0	10	35	64
	PrecStd	203	90	78	52	42	31	23	16	32	41	70	88	81
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	14.6	20.4	25.2	28.8	32.0	35.0	33.0	30.1	22.2	16.4	13.3
		MCWB	11.7	9.6	12.4	14.9	16.3	18.7	21.0	19.5	17.9	14.0	13.8	11.9
	2%	DB	11.7	12.7	16.3	20.8	25.5	28.3	31.3	30.4	27.2	19.3	14.1	11.3
		MCWB	10.1	9.4	10.6	12.6	15.8	17.3	19.1	18.7	17.2	13.5	12.3	10.2
	5%	DB	10.4	11.2	13.9	17.6	22.5	25.2	28.6	28.2	24.6	17.2	12.8	9.9
		MCWB	9.2	8.7	9.5	11.3	14.5	16.3	18.5	18.3	16.5	13.2	11.4	8.8
	10%	DB	9.4	9.9	12.0	15.0	19.6	22.3	26.1	25.9	22.2	15.5	11.6	8.7
		MCWB	8.2	7.9	8.9	10.0	13.3	15.1	17.4	17.5	15.6	12.6	10.2	7.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	13.3	15.5	17.8	19.7	21.8	20.7	19.6	16.6	14.6	12.3
		MCDB	13.2	13.2	17.7	23.8	25.7	30.2	33.2	29.7	27.7	19.1	16.0	13.1
	2%	WB	10.6	10.4	11.6	13.4	16.4	18.1	20.0	19.7	18.1	15.0	12.8	10.4
		MCDB	11.2	11.6	14.8	19.4	24.1	26.6	29.5	28.7	24.8	17.6	13.8	11.1
	5%	WB	9.4	9.1	10.5	11.9	15.1	16.7	18.8	18.7	17.1	13.9	11.7	9.0
		MCDB	10.2	10.5	13.1	16.2	21.4	24.0	27.6	27.0	22.9	16.0	12.6	9.8
	10%	WB	8.4	8.0	9.5	10.7	13.8	15.6	17.8	17.8	16.1	12.9	10.5	7.8
		MCDB	9.3	9.5	11.8	14.1	18.9	21.5	25.4	25.1	21.1	14.8	11.4	8.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	8.4	9.6	10.8	11.8	12.0	14.1	14.2	13.2	10.0	7.6	6.1
		MCDBR	7.3	11.0	13.8	17.0	18.7	18.7	19.7	19.4	18.6	13.8	8.4	6.3
	5% WB	MCWBR	5.5	7.7	8.3	9.2	9.3	8.7	8.8	8.8	9.5	8.5	6.2	5.2
		MCWBR	5.7	8.0	11.1	13.8	15.6	16.3	17.9	17.5	15.4	10.3	6.6	5.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.328	0.315	0.332	0.357	0.380	0.366	0.351	0.355	0.342	0.333	0.328	0.331
		taud	2.473	2.544	2.509	2.409	2.362	2.425	2.487	2.478	2.520	2.544	2.497	2.407
	Ebn at Noon	Ebn at Noon	760	864	896	898	885	898	907	890	874	828	754	700
		Edh at Noon	67	75	90	109	118	112	104	100	88	74	65	64
(44) All-Sky Solar Radiation	RadAvg	1.03	1.88	2.79	4.09	5.18	5.67	6.33	5.42	4.00	2.28	1.22	0.85	
	RadStd	0.16	0.26	0.35	0.39	0.44	0.57	0.48	0.39	0.37	0.26	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page